

SOFT

Self-Supervised Optical Flow Tuning

Per-Pixel Quality Estimation Without Ground Truth

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What is Optical Flow?

A **segmentation-free** method of tracking pixel motion. Minimize energy with two components:

$$E(u, v) = \sum \rho_D(I_1(x, y) - I_2(x+u, y+v)) + \lambda \sum \rho_S(\nabla u, \nabla v)$$

Data term: brightness constancy — pixels maintain intensity as they move

Smoothness term: spatial regularization — neighboring pixels move similarly

Applications

- Video stabilization
- Object tracking
- Action recognition
- Autonomous vehicles
- Video compression
- Motion segmentation

Advantages

- Dense motion field — every pixel gets a velocity
- No object detection or segmentation required
- Works with arbitrary textures and objects
- Foundation for higher-level motion analysis

The Problem

No ground truth at deployment

Optical flow algorithms are tuned offline on benchmarks, but deployed on arbitrary content with no way to measure quality.

Scene changes invalidate tuning

Parameters optimal for one scene may fail catastrophically on another. Different regions need different configurations.

Silent failures

Without quality estimation, bad optical flow propagates downstream undetected — affecting tracking, segmentation, and reconstruction.

SOFT Approach

1. Sweep Configurations

Run optical flow with multiple parameter settings (window sizes, pyramid levels, iterations)

2. Compute Quality Metrics

For each config, compute per-pixel photometric error and 3 flow errors: consistency, traction, and sensitivity

3. Select Best

Per-pixel: choose config with lowest error. Per-image: choose config with best median.

Blind Config (per-image)

Best single config based on metrics: either photometric error or the product of photometric and mean of the three flow errors.

Ensemble (per-pixel)

Best config at each pixel. Adapts to scene — smooth regions vs boundaries.

No ground truth required — all metrics computed from image pair alone

Quality Metrics

Photo

Log-RGB photometric error after warping

$$E = \sqrt{(\log R_1 - \log R_2)^2 + (\log G_1 - \log G_2)^2 + (\log B_1 - \log B_2)^2}$$

- **Log-space:** robust to illumination changes
- **RGB:** detects color errors invisible to grayscale algorithms

Flow Errors

- **Consistency:** forward vs backward flow difference
- **Traction:** track translation perturbations in frame A
- **Sensitivity:** detect translation perturbations in frame B

Why Photo works

Under brightness constancy, photometric error scales with flow error wherever image gradients exist.

Why Flow Errors help

Flow metrics measure algorithmic stability. For stable algorithms, they reduce noise in per-pixel selection.

Key insight

Optimal metric depends on algorithm. Farneback: Photo wins. DIS: Product wins.

Farneback Results (96 configurations)

Sequence	Oracle	Best	Blind Config		Ensemble	
			Photo	Product	Photo	Product
Dimetrodon	0.51	0.61	0.61 (7)	0.92 (91)	0.60 (-1%)	0.74 (+21%)
Grove2	0.24	0.42	0.53 (89)	0.53 (90)	0.34 (-20%)	0.45 (+6%)
Grove3	0.57	1.05	1.07 (13)	1.10 (28)	0.93 (-12%)	1.05 (-1%)
Hydrangea	0.27	0.45	0.46 (15)	0.47 (31)	0.38 (-16%)	0.46 (+4%)
RubberWhale	0.18	0.32	0.32 (27)	0.32 (29)	0.24 (-22%)	0.28 (-10%)
Urban2	0.36	0.95	1.43 (38)	1.84 (47)	1.10 (+16%)	1.75 (+85%)
Urban3	1.17	2.19	2.37 (22)	3.29 (56)	2.14 (-3%)	3.20 (+46%)
Venus	0.84	1.26	1.56 (75)	1.39 (60)	1.16 (-8%)	1.31 (+4%)
Average	0.52	0.91	1.04 (36)	1.23 (54)	0.86 (-5%)	1.16 (+27%)

Photo Ensemble beats Best Config by 5% — without ground truth

Blind Config: EPE (rank/96) · Ensemble: EPE (% vs Best)

DIS Results (96 configurations)

Sequence	Oracle	Best	Blind Config		Ensemble	
			Photo	Product	Photo	Product
Dimetrodon	0.07	0.12	0.12 (41)	0.13 (43)	0.13 (+11%)	0.12 (+3%)
Grove2	0.10	0.15	0.15 (10)	0.15 (8)	0.18 (+21%)	0.15 (+3%)
Grove3	0.41	0.55	0.59 (12)	0.57 (6)	0.65 (+17%)	0.57 (+4%)
Hydrangea	0.12	0.17	0.18 (6)	0.18 (6)	0.20 (+18%)	0.18 (+4%)
RubberWhale	0.07	0.10	0.10 (8)	0.10 (7)	0.15 (+45%)	0.10 (+0%)
Urban2	0.25	0.41	0.41 (1)	0.41 (1)	0.53 (+28%)	0.38 (-9%)
Urban3	0.84	1.40	1.64 (13)	1.84 (42)	1.68 (+20%)	1.42 (+1%)
Venus	0.20	0.30	0.31 (21)	0.31 (18)	0.35 (+18%)	0.30 (+1%)
Average	0.26	0.40	0.44 (14)	0.46 (16)	0.48 (+20%)	0.40 (+1%)

Product Ensemble matches Best Config (+1%) — Photo fails (+20%)

Different algorithm, different metric. DIS is inherently stable — per-pixel Photo adds noise.

Conclusions

1. Self-supervised quality estimation with per-pixel uncertainty

Blind Config selection ranks in top 40% without any labels.

2. Ensemble flows match or beat Best Config

Photo Ensemble: -5% for Farneback. Product Ensemble: +1% for DIS. All without ground truth.

3. Ensemble approach suits heterogeneous scenes

Large windows for smooth flow, small windows for boundaries. Per-pixel selection captures this automatically.

4. Computational cost is 10-50× single config

However, config evaluations are independent and embarrassingly parallel — wall-clock approaches single-config with sufficient cores.

SOFT

Know when to trust your optical flow —
without ground truth, without training, at runtime.

-5%

vs Best Config

Farneback Photo

+1%

vs Best Config

DIS Product

Top 40%

Blind Config

Without GT

Appendix: Sun et al. (2010)

Key Secrets Adopted

- **Image preprocessing:** Normalize/filter images
- **Median filtering:** 5×5 after each warp iteration
- **Coarse-to-fine:** Multi-resolution pyramid
- **Texture decomposition:** ROF separation
- **Bicubic interpolation:** Spline-based warping
- **Graduated non-convexity:** Quadratic → robust
- **Generalized Charbonnier:** $\rho(x) = (x^2 + \epsilon^2)^{0.45}$

The Paradox

Median filtering **increases energy** but **improves accuracy**. Led to new objective with non-local term.

Drawbacks

- **Computational cost:** ~70 min/image
- **Weighted non-local term:** Complex implementation
- **No per-pixel quality:** Single config for image

Appendix: Quality Metrics — Detailed Formulations

Photometric Error

Log-RGB residual after warping:

$$E = \sqrt{[\sum (\log I_A^c - \log I_B^c(x+u))^2]}$$

Log-space: robust to illumination. RGB: detects color errors.

Flow Errors

- Consistency:** $|u_{\text{fwd}}(x) + u_{\text{bwd}}(x + u_{\text{fwd}})|$
- Traction:** $|u(A_{\text{shifted}}, B) - u(A, B) - \delta| / |\delta|$
- Sensitivity:** $|u(A, B_{\text{shifted}}) - u(A, B) - \delta| / |\delta|$

Normalization: Pollution Depth

$$\text{norm} = \text{pollution_depth}^2 + \text{displacement}^2$$

Area compensation: Pollution depth is the smoothing radius. Squaring gives smoothing area — multiply to compensate for spatial averaging.

Product Metric

$$E_{\text{prod}} = E_{\text{photo}} \times \text{mean}(\text{consistency}, \text{traction}, \text{sensitivity})$$

Photometric is primary accuracy proxy. Flow errors provide stability weighting.

Appendix: Understanding Sensitivity and Texture

The Aperture Problem

In regions with little texture or color variation, optical flow is underconstrained. Algorithm can only estimate motion perpendicular to local gradients.

Why Larger Windows Help

In low-texture patches:

- **Small window:** Insufficient texture → erratic response
- **Large window:** Grabs surrounding texture → stable response

Sensitivity correctly identifies this — larger sampling = better flow in homogeneous regions.

The Tradeoff at Boundaries

Same large windows cause problems at motion boundaries:

- **Low texture:** Large window optimal ✓
- **Motion boundary:** Small window optimal (avoid bleeding) ✗

Practical Implication

Sensitivity measures stability, not accuracy. Works best as **complement** to photometric:

- **Photometric:** accurate where gradients exist
- **Sensitivity:** stable in low-texture regions
- **Product:** combines both signals

Appendix: RAFT (2020)

Key Innovations

- All-pairs correlation:** 4D cost volume for all pixel pairs
- Recurrent refinement:** GRU iteratively updates flow
- Multi-scale lookups:** Correlation pyramid pooling
- No coarse-to-fine:** Single resolution, iterative
- Strong generalization:** Cross-dataset transfer

Results

- KITTI:** 5.10% F1-all (16% improvement)
- Sintel:** 2.86 EPE (30% improvement)
- Current state-of-the-art architecture

Supervised Training

Trained on synthetic datasets (FlyingChairs, FlyingThings3D) with ground truth flow. Fine-tuned on target benchmarks.

SMURF Extension (2021)

Unsupervised RAFT using photometric + bidirectional consistency losses. 36-40% improvement over prior unsupervised methods.

Relation to SOFT

RAFT is a single trained network; SOFT selects among classical algorithm configurations per-pixel. Complementary approaches for different use cases.

Teed, Z. & Deng, J. "RAFT: Recurrent All-Pairs Field Transforms for Optical Flow." ECCV 2020.

Stone, A. et al. "SMURF: Self-Teaching Multi-Frame Unsupervised RAFT with Full-Image Warping." CVPR 2021.

operating envelope for half-plane sprite

